Fast and Calibrated: Amortized Simulation-Based Inference for Log-Gaussian Cox Processes

Elizaveta Semenova

Department of Epidemiology and Biostatistics, Imperial College London

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Abstract

Log-Gaussian Cox processes (LGCPs) are the default model for aggregated spatial count and point-pattern data in ecology, epidemiology and the social sciences, but Bayesian inference for their parameters is costly: every dataset requires a fresh Markov chain Monte Carlo (MCMC) run that must repeatedly integrate over a high-dimensional latent Gaussian field. *Amortized* simulation-based inference (SBI) removes this per-dataset cost while preserving the full Bayesian posterior, and recent work has shown it is feasible for LGCPs; the open question is whether the resulting posteriors can be *trusted*. We answer it with a systematic calibration audit. We train a neural posterior estimator—a convolutional summary network feeding a masked autoregressive flow—on simulations from a grid-discretized LGCP, and evaluate it with the field's most demanding diagnostics. The learned posterior passes simulation-based calibration on all three parameters (baseline intensity, spatial variance and correlation range) and attains near-nominal credible-interval coverage. The common point-estimate "regression" approach, by contrast, fails calibration in a way that only simulation-based calibration exposes: its intervals are right on average yet uninformative per dataset—their width barely varies and does not track the actual error—so they cannot flag which estimates to trust. Against a gold-standard No-U-Turn Sampler, the amortized posteriors are statistically indistinguishable—a classifier two-sample test accuracy of 0.543 (chance = 0.5)—yet are produced in 12.67 ms per dataset versus 280 s for MCMC, a per-dataset speed-up of $\sim 22,123\times$ that repays its one-time training cost after only 5 datasets. Finally, an honest kernel misspecification study shows the method degrades gracefully rather than failing silently. Amortized SBI thus makes calibrated, likelihood-free LGCP inference practical at the scale of modern spatial data.

1 Introduction

Spatial point patterns—the locations of trees, disease cases, crimes, or animal sightings—are ubiquitous, and the *log-Gaussian Cox process* (LGCP) is their canonical model [Møller et al., 1998, Diggle et al., 2013]. An LGCP is a doubly stochastic Poisson process whose log-intensity surface is a Gaussian process (GP); its parameters—overall abundance, the strength of spatial aggregation, and the spatial scale over which that aggregation operates—are exactly the quantities ecologists and epidemiologists want to estimate and compare across regions, species or time periods.

Inference is the bottleneck. The LGCP likelihood is intractable because it marginalizes a latent GP over the whole domain; in practice one discretizes the domain into a fine grid and treats the per-cell log-intensities as a high-dimensional latent Gaussian vector [Møller et al., 1998]. Fully Bayesian inference then proceeds by MCMC over this vector jointly with the covariance parameters, which is slow and delicate because the Gaussian field and the covariance hyperparameters are strongly coupled [Christensen et al., 2006, Murray et al., 2010]. Deterministic approximations such as INLA with an SPDE representation [Rue et al., 2009, Lindgren et al., 2011] accelerate this enormously but are approximate, tied to Gaussian-Markov structure, and still solved *per dataset*. In modern applications—disease mapping across thousands of administrative units, species distribution models over many taxa, or streaming crime data—the same model is fit again and again to structurally identical but numerically distinct datasets. Paying a full inference cost every time is wasteful.

Amortized simulation-based inference offers a different trade [Cranmer et al., 2020, Zammit-Mangion et al., 2025]. One trains a neural conditional density estimator $q_\phi(\boldsymbol{\theta} \mid \mathbf{x})$ on pairs $(\boldsymbol{\theta}, \mathbf{x})$ drawn from the prior and the simulator, so that a single forward pass returns an approximate posterior for any new dataset $\mathbf{x}$. The cost of inference is moved *up front* and *shared* across all future datasets, and the method never evaluates the likelihood—only simulates from it—so it extends naturally to observation processes (thinning, preferential sampling, aggregation) that break analytic approaches.

This idea has already reached the LGCP: Wang et al. [2026] apply the BayesFlow amortized estimator [Radev et al., 2020] to one- and two-dimensional LGCPs and validate it against MCMC, and a broader line of work uses neural networks to estimate spatial point-process parameters as point predictions [Vihrs, 2022]. Feasibility, in other words, is established. The question we take up is the next one, and the one that decides whether such estimators can be *used*: are the amortized posteriors *trustworthy*? Concretely—do they pass the field's strongest calibration diagnostics across the whole prior, not just agree with MCMC on a handful of cases; how do they compare with the prevailing point-estimate paradigm; what does the choice of summary statistics cost in calibration; and how do they behave when the simulator is misspecified? This paper is a systematic answer to those questions.

Contributions. Building on the feasibility established by prior amortized LGCP work [Wang et al., 2026], our contribution is a rigorous validation and a set of controlled comparisons, using a transparent, dependency-light neural posterior estimator (a convolutional summary network feeding a masked autoregressive flow over the three interpretable parameters; Section 3):

1. We subject the learned posterior to the strongest calibration diagnostics in the SBI toolkit—simulation-based calibration over the full prior, plus marginal coverage and an *adaptive-uncertainty* analysis—and show it passes on all three parameters (Section 4).

2. We run a controlled comparison against the prevailing point-estimate paradigm [Vihrs, 2022]: a regression network matches the NPE on point-estimate accuracy yet is badly miscalibrated in a way that marginal coverage alone would miss, because its uncertainty does not adapt to the data (Section 4). We further ablate learned versus hand-crafted spatial summary statistics.

3. We benchmark against a gold-standard No-U-Turn Sampler in two dimensions: the amortized posteriors are statistically indistinguishable from MCMC (classifier two-sample accuracy 0.543) at a $\sim 22,123\times$ lower per-dataset cost, breaking even after 5 datasets (Section 5).

4. We report an honest robustness study under kernel misspecification, showing graceful rather than catastrophic degradation (Section 6).

All code and experiments are released to reproduce every number and figure.

2 Background

Log-Gaussian Cox processes. A Cox process on a domain $D \subset \mathbb{R}^2$ is a Poisson process with a random intensity $\Lambda(s)$. In an LGCP, $\log \Lambda(s) = Z(s)$ is a Gaussian process, $Z \sim \mathcal{GP}(\mu, k)$ [Møller et al., 1998]. Discretizing D into a regular $G \times G$ grid of cells with centroids s_i and equal area a, the standard computational model [Diggle et al., 2013] takes

$$Z_i = \beta_0 + f_i, \quad f \sim \mathcal{N}(\mathbf{0}, K_{\boldsymbol{\theta}}), \quad K_{ij} = \sigma^2 \rho_\nu\big(\|s_i - s_j\|/\ell\big), \quad y_i \mid Z \sim \mathrm{Poisson}\big(a\, e^{Z_i}\big), \tag{1}$$

where ρ_ν is a Matérn correlation of smoothness ν (we use the exponential kernel $\nu = \frac{1}{2}$). The parameters $\boldsymbol{\theta} = (\beta_0, \sigma, \ell)$ are the baseline log-intensity (abundance), the marginal standard deviation of the log-intensity field (aggregation / over-dispersion), and the correlation length (spatial scale of aggregation). The latent vector $f \in \mathbb{R}^{G^2}$ is a nuisance that must be integrated out; this is the source of both the modelling power and the computational difficulty.

Neural posterior estimation. Given a prior $p(\boldsymbol{\theta})$ and an implicit simulator $p(\mathbf{x} \mid \boldsymbol{\theta})$, NPE trains a conditional density estimator by minimizing the forward Kullback–Leibler divergence,

$$\phi^\star = \arg\min_{\phi} \; \mathbb{E}_{\boldsymbol{\theta}\sim p(\boldsymbol{\theta})}\, \mathbb{E}_{\mathbf{x}\sim p(\mathbf{x}\mid\boldsymbol{\theta})} \big[-\log q_\phi(\boldsymbol{\theta} \mid \mathbf{x}) \big], \tag{2}$$

whose minimizer is the true posterior $p(\boldsymbol{\theta} \mid \mathbf{x})$ when the family is flexible enough [Papamakarios and Murray, 2016, Greenberg et al., 2019]. We use a single-round (fully amortized) estimator so that, after training, inference on a new dataset is one forward pass. The density family is a masked autoregressive flow (MAF) [Papamakarios et al., 2017], and the conditioning $\mathbf{x}$ is a learned summary of the count grid.

3 Method

Simulator and prior. We work on the unit square discretized into a 16×16 grid (256 cells). The prior is $\beta_0 \sim \mathcal{N}(5, 0.7^2)$, $\sigma \sim \mathrm{U}(0.2, 1.8)$, $\ell \sim \mathrm{U}(0.05, 0.5)$, chosen so that prior-predictive patterns span sparse-to-dense and weakly-to-strongly aggregated regimes (Figure 1). Simulating from (1) requires, per parameter draw, a Cholesky factor of the 256×256 covariance; we batch these on the CPU so that $60,000$ datasets are produced in about a minute.

Parameter transform. So that posterior samples always respect the (bounded) prior support, the flow operates on an unconstrained, standardized reparameterization: β_0 is z-scored and σ, ℓ are mapped through a logit of their rescaled uniform range and standardized. Densities and samples are transformed back to natural space with the exact change of variables.

Summary network. The count grid is passed through a small convolutional network (two 3×3 conv blocks, average pooling, two further conv blocks, global pooling) whose output is concatenated with two explicit global statistics (log total count and log count-variance) and projected to a 48-dimensional embedding. The convolutions exploit the spatial structure of the pattern, while the explicit total-count feature gives the network direct access to the strongest signal for abundance.

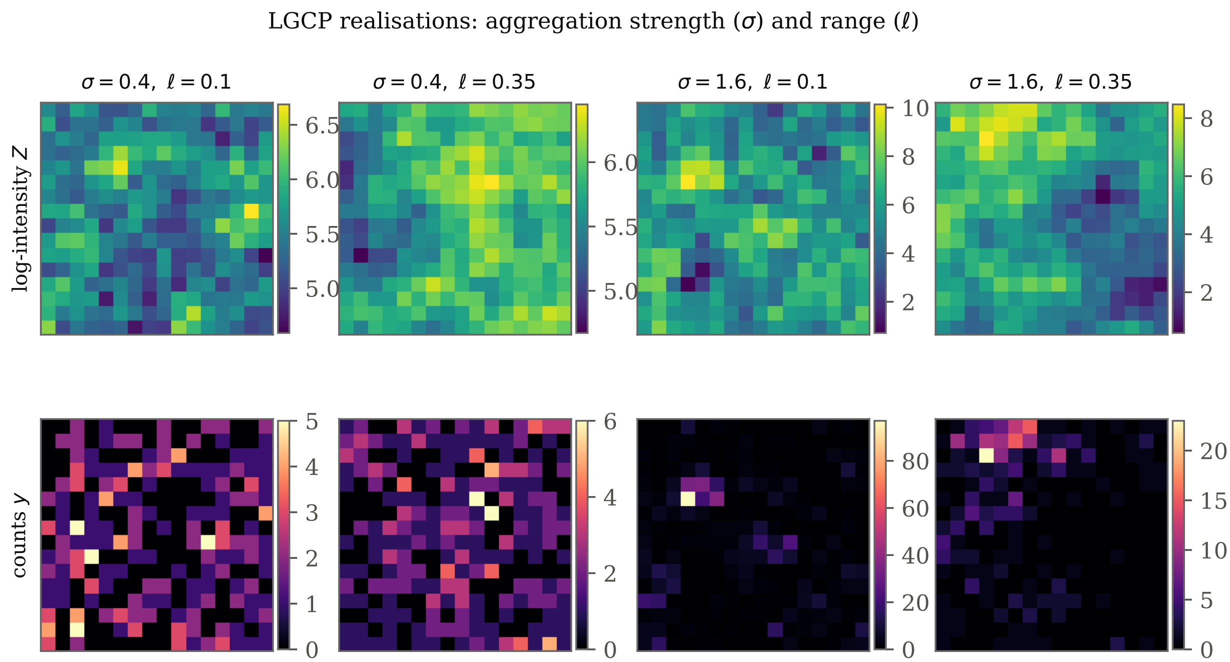

Figure 1: Prior-predictive LGCP realisations on the 16×16 grid. Top: latent log-intensity fields Z; bottom: the observed Poisson counts y. Increasing σ (left→right pairs) strengthens aggregation; increasing ℓ enlarges the spatial scale of that aggregation. Inference must recover these parameters from a *single* count grid.

Density estimator. The embedding conditions a MAF with five autoregressive transforms (each a conditional MADE with two hidden layers of width 64 and tanh-bounded log-scales for stability), with the variable order reversed between transforms. The whole model (104,398 parameters) is trained end-to-end by minimizing (2) on the $60,000$ simulations with Adam, early stopping on a validation split; training takes 22.1 minutes on four CPU cores. At test time we draw $1,000$ posterior samples per dataset.

Baselines. We compare against (i) an otherwise identical NPE that consumes *hand-crafted* spatial summaries—global count statistics, the index of dispersion, and a binned spatial autocorrelation profile of the log-count field—instead of the learned CNN embedding; and (ii) a *point-estimate CNN regressor* trained by mean-squared error, representing the common practice of "learning to invert" a simulator. To place the regressor on the same calibration axes we equip it with a homoscedastic Gaussian predictive whose scale is its training residual standard deviation.

Gold-standard reference. As a reference posterior we run the No-U-Turn Sampler (NUTS) [Hoffman and Gelman, 2014] on the *same* model and priors, using a non-centred parameterization $f = L_{\boldsymbol{\theta}}\,\eta$, $\eta \sim \mathcal{N}(\mathbf{0}, I)$, where the Cholesky factor $L_{\boldsymbol{\theta}}$ is rebuilt at every leapfrog step because it depends on the sampled length-scale ℓ. This per-step re-factorization is exactly the cost that amortization avoids. We run 2 chains of 600+500 iterations and monitor split-$\hat{R}$.

4 Calibration and recovery

We evaluate on $3,000$ held-out datasets drawn from the prior predictive.

Table 1: Held-out accuracy and calibration ($3,000$ datasets). Point-estimate accuracy (RMSE) and marginal 90% coverage are similar across methods; the minimum SBC p-value (over the three parameters) separates them. The CNN+MAF NPE passes on all parameters ($\min p > 0.05$); the hand-crafted-summary NPE is close but mildly miscalibrated; the regressor fails decisively ($\min p \approx 0$).

	RMSE ↓			90% coverage			SBC
Method	β_0	σ	ℓ	β_0	σ	ℓ	$\min p$
NPE (CNN+MAF)	0.379	0.190	0.103	0.90	0.90	0.89	0.114
NPE (hand-crafted)	0.399	0.200	0.107	0.89	0.90	0.89	0.009
CNN regressor	0.379	0.192	0.104	0.89	0.92	0.90	0.000

Simulation-based calibration. SBC [Talts et al., 2018] exploits the fact that, if q_ϕ equals the true posterior, the rank of each true parameter within its posterior sample is uniform. Figure 2 shows the NPE rank empirical CDFs for all three parameters lying inside the 95% uniformity band, with χ^2 uniformity p-values 0.64/0.15/0.11 (all > 0.05): the amortized posterior is calibrated in the strong distributional sense SBC demands. The point-estimate regressor's ranks (orange) depart drastically from uniform ($p < 0.01$ on every parameter), the signature of a posterior approximation that is systematically wrong per dataset.

Marginal coverage. Figure 3 plots empirical against nominal central-interval coverage. Both NPE variants track the diagonal closely—the CNN+MAF model's 90% intervals cover 0.90/0.90/0.89 for $\beta_0/\sigma/\ell$. Strikingly, the regressor *also* appears near-nominal here (0.89/0.92/0.90). This is not a contradiction of its failed SBC: a homoscedastic predictive whose width is tuned on held-out residuals is guaranteed to cover correctly *on average over the prior.* Marginal coverage is necessary but not sufficient—and on its own it would have hidden the regressor's defect.

Adaptive uncertainty. SBC's verdict is made concrete by asking whether the reported uncertainty predicts the actual error. Figure 4 bins datasets by their reported posterior standard deviation and plots the realised RMSE in each bin. For the NPE the two rise together along the diagonal—the posterior is *sharper* on easy datasets and wider on hard ones—so its width is informative (correlation between reported SD and error 0.37/0.40/0.43 for $\beta_0/\sigma/\ell$). The regressor's homoscedastic predictive cannot do this: most starkly for β_0, whose intervals have a nearly constant width (coefficient of variation 0.02 versus 0.30 for the NPE) and zero correlation with the error ($r = 0.01$). This is why the regressor fails SBC while looking calibrated marginally, and it is the central practical warning of the paper: inverting a simulator with a regression network yields uncertainty that is uninformative per dataset, whereas NPE yields a calibrated, input-adaptive posterior at essentially the same training cost.

Recovery. Figure 5 shows posterior means against truth. Abundance β_0 and aggregation σ are recovered well; the correlation length ℓ is intrinsically harder to pin down from a single realization, and— importantly—the NPE *reports* this through wider, still-calibrated intervals rather than confident errors. Coefficients of determination are 0.70/0.83/0.37. The weaker identifiability of β_0 against the field mean (a known feature of LGCPs) is faithfully reflected in the posterior width, not hidden.

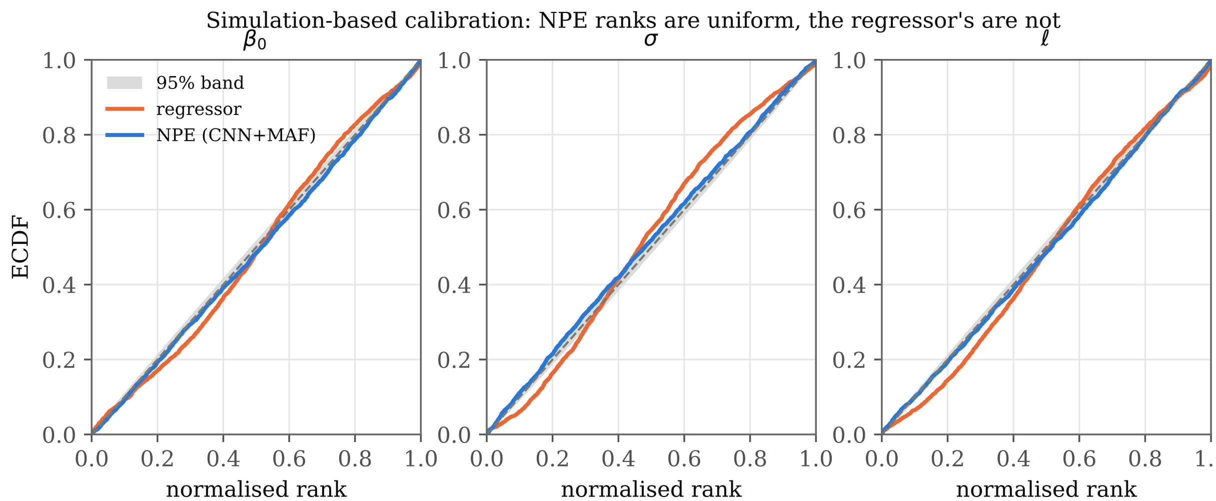

Figure 2: Simulation-based calibration. NPE rank ECDFs (blue) for the three parameters fall within the 95% uniformity band (grey); the diagonal is perfect calibration. The point-estimate regressor (orange) departs sharply from uniform, revealing per-dataset miscalibration.

Table 2: Per-parameter agreement between NPE and NUTS over 30 datasets. C2ST is computed on the joint 3-D posteriors (0.5 = indistinguishable).

Parameter	C2ST	1-Wass. (/prior sd)	mean corr	sd ratio
β_0	–	0.056	0.998	1.03
σ	–	0.053	0.998	1.04
ℓ	–	0.091	0.989	1.03
overall (3-D)	0.543	0.067	–	–

5 Agreement with gold-standard MCMC

Calibration shows the posterior is right *on average* over the prior; we now ask whether it is right *for individual datasets* by comparing, dataset by dataset, against NUTS. We ran NUTS on 30 test datasets that reached convergence ($\hat{R} < 1.1$).

Figure 6 tells the story. The overlaid marginal posteriors (top) are nearly coincident with the NUTS reference for a representative dataset. The posterior means agree tightly (correlation ≥ 0.99 across parameters) and, crucially, so do the posterior *standard deviations*—the NPE is not merely locating the posterior but reproducing its spread. Aggregated over datasets, a classifier two-sample test cannot tell NPE from NUTS samples apart: mean C2ST accuracy is 0.543 against a chance level of 0.5, and the mean marginal 1-Wasserstein distance is only 0.067 prior standard deviations. Table 2 collects these numbers.

The practical pay-off is in Figure 7. A single NUTS fit takes 280 s; an amortized NPE posterior takes 12.67 ms, a $\sim 22,123\times$ per-dataset speed-up. Accounting for the one-time 22.1-minute training cost, NPE overtakes repeated MCMC after just 5 datasets—a threshold trivially exceeded in any realistic multi-region or multi-species study.

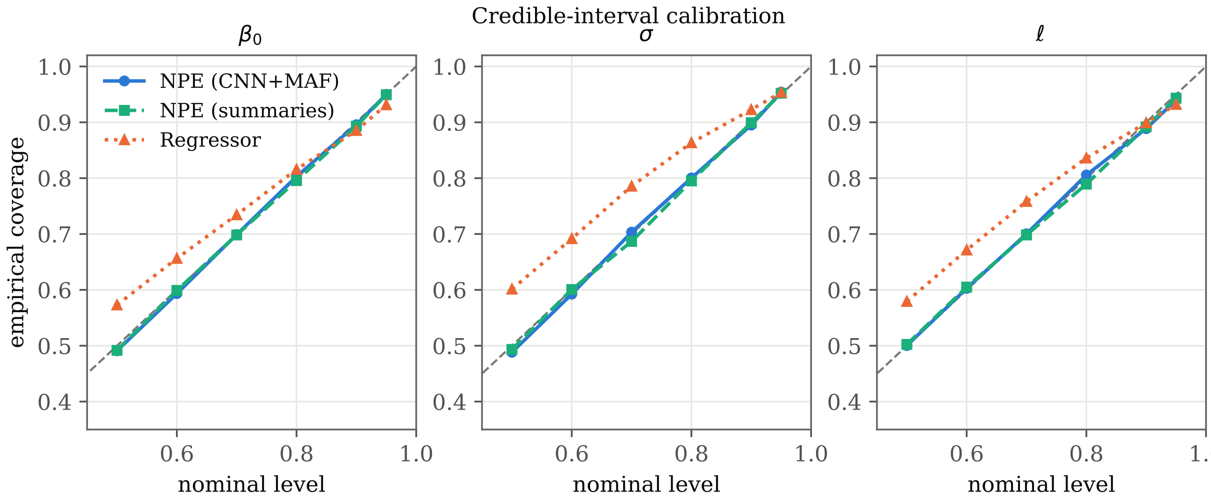

Figure 3: Marginal credible-interval calibration. All three methods track the diagonal: the regressor's tuned homoscedastic width makes it look calibrated *marginally*, which Figure 4 shows to be misleading.

6 Robustness to misspecification

A method that is only validated in-distribution can fail silently when the world does not match the simulator. We therefore train on the exponential ($\nu = \frac{1}{2}$) kernel but *test* on data generated from rougher-to-smoother Matérn kernels ($\nu = \frac{3}{2}, \frac{5}{2}$), a realistic form of model mismatch since the true smoothness is rarely known. Figure 8 shows coverage degrading gracefully: intervals remain informative and only mildly miscalibrated under moderate mismatch (at $\nu = \frac{5}{2}$, 90% coverage 0.87/0.75/0.75), with ℓ—whose meaning is most kernel-dependent —the most affected. The failure mode is gradual and diagnosable rather than catastrophic, and it is detectable in practice with the same SBC/coverage tools used above, supporting a workflow in which the simulator's smoothness is itself treated as uncertain.

7 Related work

Simulation-based inference. NPE originates with mixture-density and flow-based conditional estimators [Papamakarios and Murray, 2016], refined by sequential and automatic-posterior variants [Lueckmann et al., 2017, Greenberg et al., 2019] and surveyed by Cranmer et al. [2020]. Alternatives estimate the likelihood [Papamakarios et al., 2019] or the likelihood ratio [Hermans et al., 2020]; Lueckmann et al. [2021] benchmark them. Our density estimator is a masked autoregressive flow [Papamakarios et al., 2017, Rezende and Mohamed, 2015]. Calibration is assessed with simulation-based calibration [Talts et al., 2018] and the classifier two-sample test [Lopez-Paz and Oquab, 2017]; recent work stresses that SBI must be checked, not assumed [Hermans et al., 2022].

Inference for LGCPs and spatial models. Classical LGCP inference uses MCMC over the latent field [Møller et al., 1998, Christensen et al., 2006], with elliptical slice sampling and manifold methods to cope with the field–hyperparameter coupling [Murray et al., 2010, Girolami and Calderhead, 2011]. INLA with an SPDE representation [Rue et al., 2009, Lindgren et al., 2011, Simpson et al., 2016] is the dominant fast, approximate alternative. These are all *per-dataset* solvers;

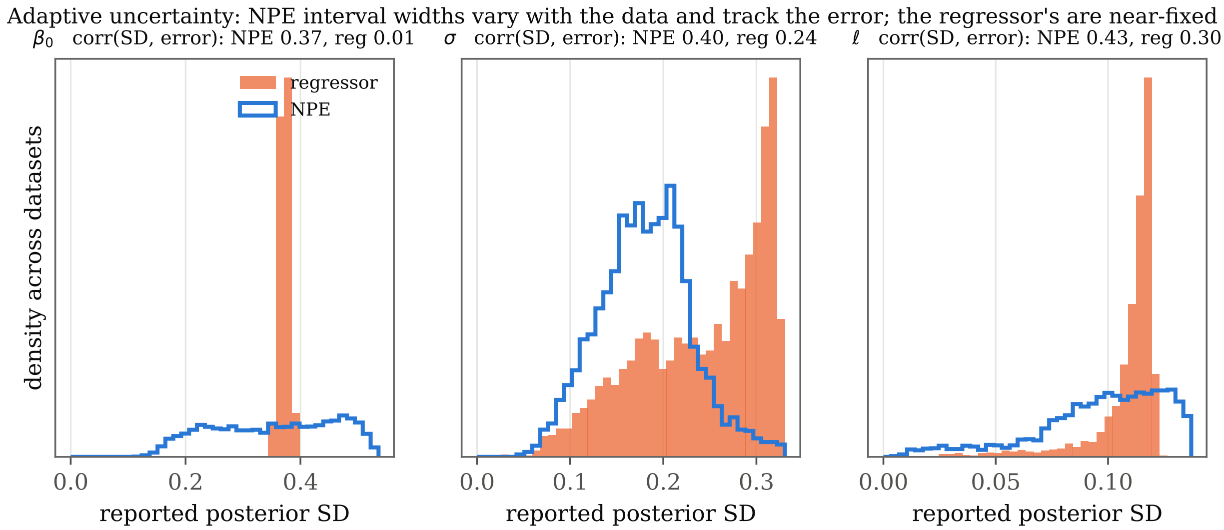

Figure 4: Does the reported uncertainty predict the error? Datasets are binned by their posterior standard deviation (x); y is the realised RMSE in each bin. NPE (blue) follows the diagonal—wider intervals where the error is genuinely larger—so its uncertainty is informative. The regressor (orange) is flat, especially for β_0: its fixed-width predictive says nothing about which estimates are reliable. This is the per-dataset miscalibration that SBC detects and marginal coverage hides.

our contribution is orthogonal—an amortized posterior shared across datasets—and complementary, since INLA or NUTS can serve as the gold standard we validate against.

Neural inference for spatial and point-process models. Amortized and likelihood-free neural inference has moved rapidly into spatial statistics and ecology; see Zammit-Mangion et al. [2025] for a review. Two threads are directly relevant. The first estimates parameters as *point predictions*: Vihrs [2022] trains neural networks to recover spatial point-process parameters from a pattern, and neural Bayes estimators provide fast, amortized point estimators for spatial covariance and extremal models, with bootstrap uncertainty [Gerber and Nychka, 2021, Lenzi et al., 2023, Sainsbury-Dale et al., 2024, 2025]. Our regressor baseline is a representative of this thread, and our calibration analysis quantifies precisely what its uncertainty does and does not deliver. The second thread learns *full posteriors*: closest to us, Wang et al. [2026] apply BayesFlow [Radev et al., 2020] to one- and two-dimensional LGCPs and validate against MCMC on selected realizations, with an oral-microbiome application. We are complementary to that work: rather than a new application, we contribute a systematic *trustworthiness audit*—calibration across the entire prior via SBC, a controlled comparison against the point-estimate paradigm, a summary-statistic ablation, a classifier-two-sample and Wasserstein benchmark against MCMC in two dimensions, and a kernel-misspecification stress test—using a deliberately minimal, from-scratch estimator so that the conclusions are about the approach, not a particular software stack.

8 Discussion

We have shown that amortized NPE gives calibrated posteriors for LGCP parameters that match gold-standard MCMC at a fraction of the per-dataset cost, and that the common regression shortcut, while accurate in the mean, is dangerously overconfident. The recipe is simple and dependency-

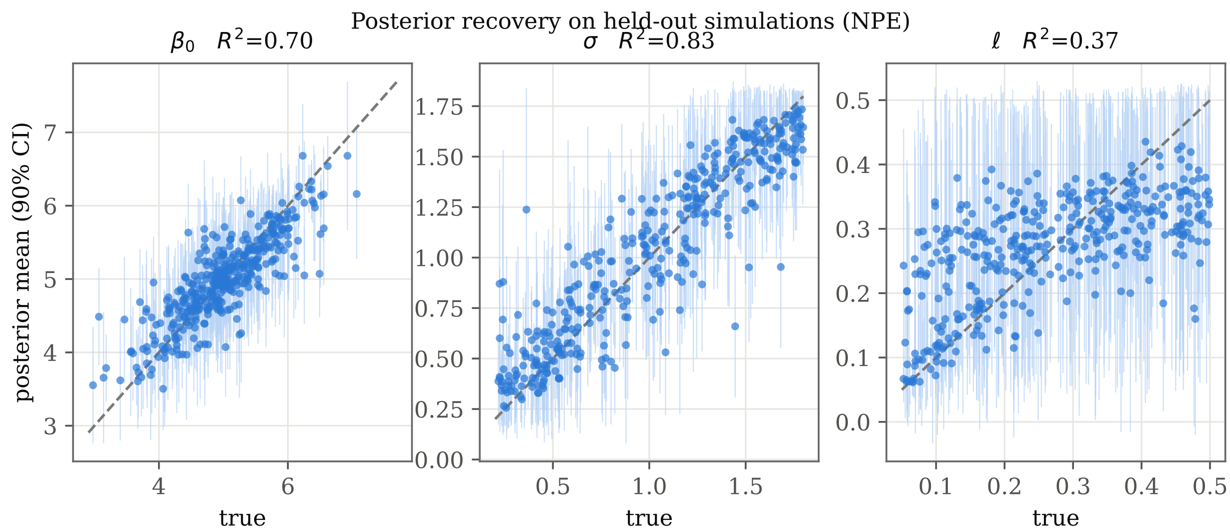

Figure 5: Posterior recovery on held-out simulations. Points are posterior means with 90% credible intervals against the true parameter; the dashed line is $y = x$.

light: simulate, embed with a CNN, fit a flow, and—critically—*check* with SBC, coverage and a gold-standard comparison before trusting the result.

Limitations and outlook. Our study fixes the grid resolution and the kernel family, infers three global parameters, and does not reconstruct the latent field; the misspecification analysis shows real sensitivity to smoothness. Natural extensions are to amortize over grid resolution and domain shape, to jointly infer the latent intensity surface (a spatial prediction task), to add covariates and non-stationarity, and to handle realistic observation processes—thinning, preferential sampling and aggregation—where the likelihood-free nature of SBI is decisive and analytic competitors do not apply. Treating the kernel smoothness as an inferred parameter, rather than a fixed modelling choice, is a promising route to robustness. We view calibrated amortized inference as a practical path to LGCP modelling at the scale that contemporary spatial data demand.

Reproducibility. The simulator, method, diagnostics and all experiment scripts are released; every figure and number in this paper is regenerated by the accompanying code.

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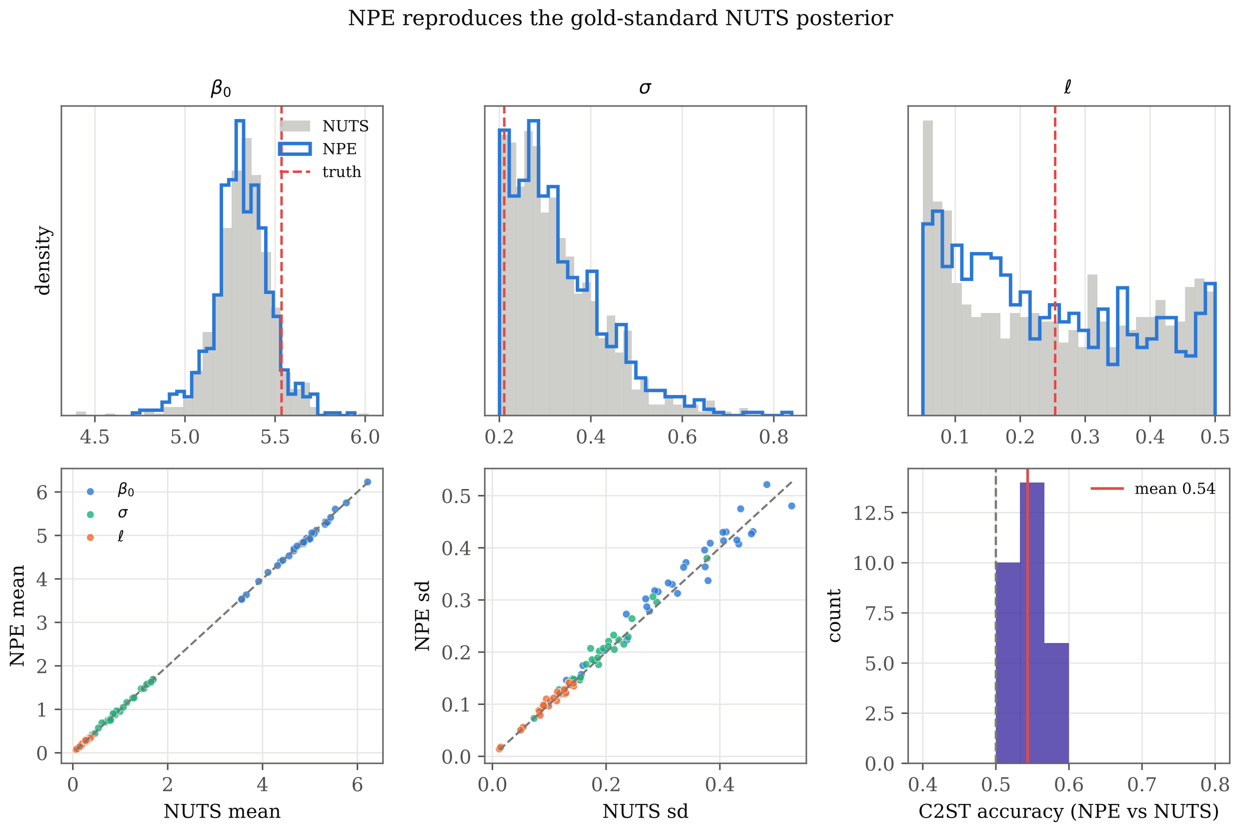

Figure 6: NPE versus gold-standard NUTS. *Top:* overlaid marginal posteriors for one held-out dataset (NPE step outline, NUTS filled, truth dashed). *Bottom:* across datasets, posterior means and standard deviations agree with NUTS, and the classifier two-sample accuracy concentrates near the chance level 0.5 (indistinguishable posteriors).

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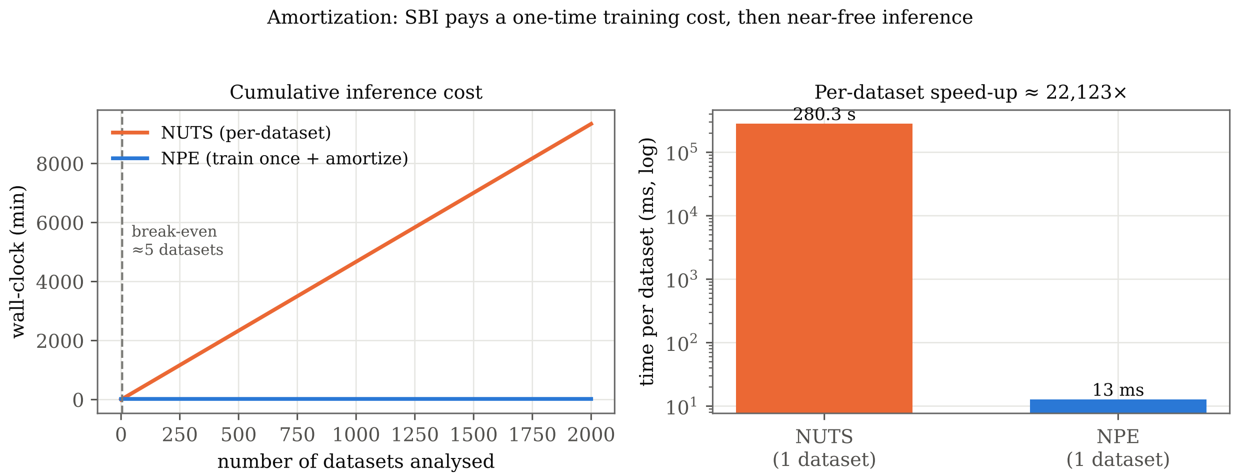

Figure 7: Amortization. *Left:* cumulative wall-clock against number of datasets; the NPE line (training cost + near-free inference) crosses the per-dataset MCMC line after 5 datasets. *Right:* per-dataset cost on a log scale.

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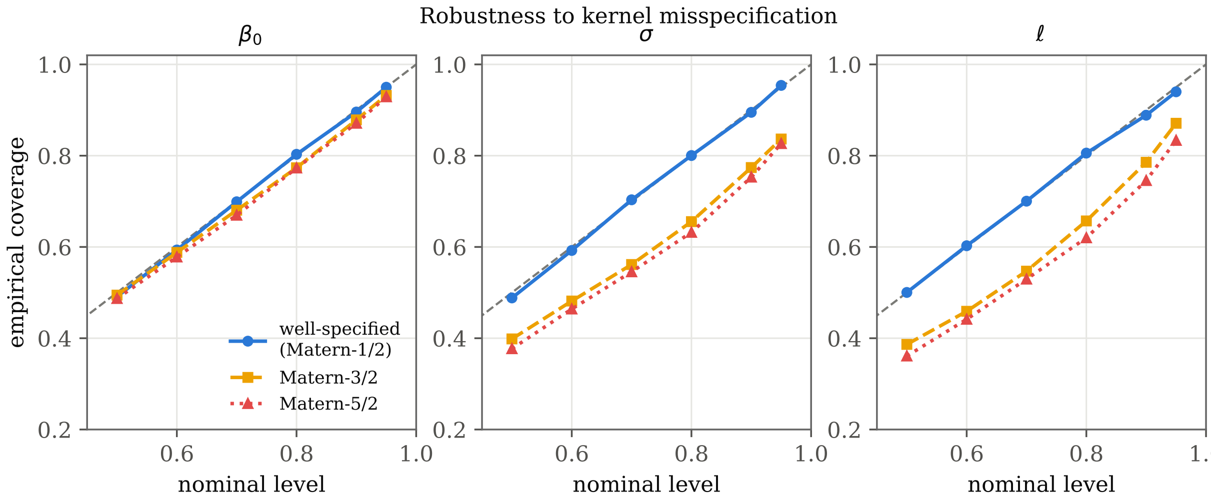

Figure 8: Kernel misspecification. Coverage of the NPE (trained on $\nu = \frac{1}{2}$) under test data from $\nu = \frac{1}{2}$ (well-specified), $\frac{3}{2}$ and $\frac{5}{2}$. Degradation is gradual and largest for the range ℓ.

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